Anjuman-i-Islam's
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8, Saboo Siddik Polytechnic Road, Byculla, Mumbai-8

IoT Mini Project - Report Format

Department of Computer Science and Engineering (IOTCSBCT) | A.Y. 2026-2027

Project requirement

Build a system based on **one real-life application** that uses a **minimum of 2–3 sensors**, **at least 1 actuator**, and **cloud connectivity** (data sent to and viewed on a cloud platform or dashboard).

Weightage: 10 marks out of 25 Term Work marks (Experiments 10 + Mini Project 10 + Attendance 5). The external oral will be based on the 10 experiments and your mini project.

1. Report Structure

#	Section	What to include
–	**Title Page**	Project title, group members' names and roll numbers, batch, subject in-charge, department, college, academic year (see sample on page 3).
–	**Acknowledgement**	Short note thanking the guide, department and anyone who helped.
–	**Abstract**	150–250 words: the problem, sensors and actuator used, cloud platform, and the outcome.
–	**Table of Contents & List of Figures**	Chapter titles, figure/table titles with page numbers.
1	**Introduction**	Background of the real-life problem, problem statement, objectives, and scope of the project.
2	**Literature Survey / Existing Systems**	2–3 existing solutions or papers, their limitations, and how your project improves on them.
3	**System Design**	Block diagram, components table (see Section 2 below), circuit diagram / pin connections, and a flowchart of the system logic.
4	**Implementation**	Hardware setup, software tools (IDE, libraries), cloud platform setup (channel/dashboard/API), and key code snippets with explanation.
5	**Results**	Photos of the working prototype, cloud dashboard screenshots, table of sensor readings, and how the actuator responds.
6	**Conclusion & Future Scope**	What was achieved, limitations, and possible improvements.
–	**References**	At least 5, in IEEE format, e.g. [1] A. Author, "Title," *Journal*, vol., no., pp., year.
–	**Appendix**	Complete source code and component datasheet links.

2. Components Table (to be included in Chapter 3)

Students must list every hardware and cloud component in this format. The table must show at least 2 sensors, 1 actuator and 1 cloud platform.

Sr. No.	Component	Category	Model / Specification	Purpose in Project
1	*e.g. Temperature & humidity sensor*	Sensor	*DHT11*	*Monitors room conditions*
2	*e.g. Soil moisture sensor*	Sensor	*Capacitive v1.2*	*Detects dry soil*
3	*e.g. Water pump via relay*	Actuator	_5V relay + DC pump_	*Waters plants automatically*
4	*e.g. Microcontroller*	Controller	*ESP32 / NodeMCU*	*Reads sensors, drives actuator, sends data*
5	*e.g. Cloud platform*	Cloud	*ThingSpeak / Blynk / Firebase*	*Stores data, shows live dashboard*

3. Formatting Guidelines

Paper & margins	A4; margins 1 inch on all sides (1.5 inch left if binding).
Font	Times New Roman – body 12 pt, chapter headings 16 pt bold, sub-headings 14 pt bold.
Spacing	1.5 line spacing; text justified.
Figures & tables	Numbered chapter-wise (Fig. 3.1, Table 5.2) with a caption; figure captions below, table captions above.
Page numbers	Roman numerals (i, ii, iii) for front pages; Arabic (1, 2, 3) from Chapter 1 onward.
Length	Approximately 15–25 pages excluding appendix.
Screenshots	Clear and readable; crop to the relevant area. Phone photos of the prototype are fine if well lit.

4. Evaluation Rubric (10 Marks)

Criteria	What is checked	Marks
Problem & Relevance	Clear real-life problem, well-defined objectives.	1
System Design	Block diagram, circuit diagram, flowchart and components table are complete and correct.	2
Hardware Implementation	Minimum 2–3 sensors and 1 actuator integrated and working.	2
Cloud Connectivity	Live data uploaded to the cloud; dashboard / alerts / remote control demonstrated.	2
Report Quality	Follows this format, results documented with screenshots, references cited.	2
Demonstration & Viva	Working demo and each member can explain the project.	1
Total		**10**

5. Sample Title Page

A Mini Project Report on

"<PROJECT TITLE IN CAPITAL LETTERS>"

Submitted in partial fulfilment of the Term Work for
<Lab / Course Name> (Course Code: _____)

Submitted by
Name 1 (Roll No. ___) Name 2 (Roll No. ___) Name 3 (Roll No. ___)
Batch: ____ Semester: ____

Under the guidance of
<Subject In-charge Name>

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M. H. Saboo Siddik College of Engineering, Byculla, Mumbai-8
Academic Year 2026-2027